Periodic, Wold, and Mixing Structure of Integer Expanding Circle Maps

Route-A structural certificate C177

Abstract

For every integer $b \geq 2$, we solve the fixed points and Artin–Mazur zeta of $T_b(x) = bx \pmod 1$, decompose its Haar–Koopman operator into a constant and countably many unilateral shifts, and prove a sharp Sobolev correlation law. The result couples an exact primitive-orbit ledger to its natural nonunitary operator owner and exposes a degree-only Route-A obstruction.

Keywords: expanding circle map; Artin–Mazur zeta; Koopman isometry; Wold decomposition; primitive cycle; correlation decay.

中文摘要

本文对任意整数$b \geq 2$的扩张圆映射给出全部不动点、原始周期轨道与动力 ζ函数，并把自然Koopman算子精确分解为常数部分和可数无穷个 单边移位，并证明尖锐的光滑观测相关衰减律。周期数据虽然完整，却只依赖拓扑次数， 因而构成严格的Route-A反例。

关键词：扩张圆映射；动力ζ函数；Koopman等距算子； Wold分解；原始周期。

1 Periodic theorem

Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ and $T_b(x) = bx \pmod 1$. The equation $T_b^n(x) = x$ is $(b^n - 1)x = 0$ in $\mathbb{T}$, so

$$\mathrm{Fix}(T_b^n) = \left\{ \frac{j}{b^n - 1} : 0 \leq j < b^n - 1 \right\}, \qquad F_b(n) = b^n - 1.$$

Möbius inversion gives exact-period points and primitive cycles

$$P_b(n) = \sum_{d|n} \mu(n/d)(b^d - 1), \qquad C_b(n) = P_b(n)/n.$$

Consequently

$$\zeta_{AM,b}(z) = \exp \sum_{n \geq 1} \frac{(b^n - 1)z^n}{n} = \frac{1 - z}{1 - bz} = \prod_{n \geq 1} (1 - z^n)^{-C_b(n)}.$$

The product is coefficientwise. Since exact-period points split into n-cycles, the divisibility $n \mid P_b(n)$ is intrinsic, not empirical.

2 Natural operator owner

For $e_m(x) = e^{2\pi i m x}$ and $U_b f = f \circ T_b$, one has $U_b e_m = e_{bm}$. Every nonzero integer is uniquely $m = rb^j$ with $b \nmid r$, hence

$$L^2(\mathbb{T}) = \mathbb{C}1 \oplus \bigoplus_{\substack{r \neq 0 \\ b \nmid r}} \overline{\mathrm{span}}\{e_{rb^j} : j \geq 0\}, \qquad U_b \simeq 1 \oplus S^{(\aleph_0)}.$$

Each nonconstant chain is a unilateral shift. Thus the spectrum is the closed unit disk and the only eigenvalue is 1 on constants. Moreover

$$U_b^* e_m = \begin{cases} e_{m/b}, & b \mid m, \\ 0, & b \nmid m. \end{cases}$$

The source Koopman operator is a proper isometry, not a unitary: its range omits e_1. It is noncompact, belongs to no finite Schatten class, and for $z \neq 0$ has no ordinary Fredholm determinant $\det(I - zU_b)$.

Indeed the chain vectors form an infinite orthonormal sequence whose images remain orthonormal, ruling out compactness. All nonzero singular values of an isometry are one, so their p-sum diverges for every finite p. This also blocks trace-class ownership of zU_b when $z \neq 0$. The adjoint formula follows directly from $\langle e_k, U_b e_m \rangle = \mathbf{1}_{k=bm}$ and shows that the Perron operator discards precisely the modes not divisible by b.

Boundary cases. The hypothesis $b \geq 2$ is essential. At $b = 1$ the map is the identity, every fixed set is infinite, and the ordinary Artin–Mazur definition does not apply. Negative and noninteger slopes lie outside the frozen family.

3 Sharp smooth-observable correlation law

For a mean-zero f set $\|f\|_{\dot{H}^s}^2 = \sum_{k \neq 0} |k|^{2s} |\widehat{f}(k)|^2$. The Fourier action gives

$$\langle f, U_b^n g \rangle = \sum_{m \neq 0} \overline{\widehat{f}(b^n m)} \widehat{g}(m).$$

Inserting $|b^n m|^s$, using $|m|^{-s} \leq 1$, and applying Cauchy–Schwarz proves, for $s \geq 0$,

$$|\langle f, U_b^n g \rangle| \leq b^{-ns} \|f\|_{\dot{H}^s} \|g\|_2.$$

For $s > 0$ this exponent and constant are sharp: $f = e_{b^n}$ and $g = e_1$ give equality after division by their norms. Thus the same chain ledger simultaneously owns periodic growth, the Perron filter, and quantitative mixing; none of these layers distinguishes prime from composite degree.

4 Route-A decision

Gate	Verdict	Reason
A0	`FAIL`	no intrinsic arithmetic origin
A1	`WEAK`	complete but degree-only primitive ledger
A2	`FAIL`	generic rational source zeta
A3	`FAIL`	no target global comparison
A4	`FORMAL_HINT`	unitary only after inverse-limit extension

Prime and composite b satisfy the identical theorem. The overall v0.2 verdict is `ROUTE_A_REJECTED`; Route B is false. The inverse-limit unitary dilation changes phase space and is not a physical quantization.

Evidence boundary. Finite exact evidence contains 132 periodic rows, 1,595 Wold rows, and 352 sharp-correlation rows. A producer-independent checker passes 3,980 assertions, SymPy passes 3,927 checks, byte replay is exact, and all 19 hostile mutations are rejected. These are regression tests; the proof carries every all-parameter statement. Citation and reference populations are zero, and no novelty or priority claim is made. Scope: `NO_BAD_EULER_OR_ROOT_NUMBER`.

Declarations

Data and code availability. Package-local exact evidence and deterministic code accompany this manuscript.

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